

Explainable AI for Cybersecurity: Interpreting Machine Learning Decisions in Intrusion Detection Systems

Rajat Kumar* and Sadhil Madan**

Department of Information Technology, K. J. Somaiya School of Engineering, Mumbai, India

*ORCID: 0009-0004-1688-0855

Abstract—Machine-learning intrusion detection systems can classify large volumes of network traffic, but their predictions often provide little evidence that a security analyst can examine. This paper proposes a reproducible study of whether post-hoc explanations make Random Forest and XGBoost intrusion decisions easier to inspect without weakening detection performance. The study uses the official UNSW-NB15 training and test partitions for binary attack detection while retaining attack categories for error analysis. Preprocessing, feature selection, model tuning, and threshold selection are restricted to training data. Accuracy, precision, recall, F1-score, false-positive rate, ROC-AUC, and PR-AUC describe detection quality. SHAP and LIME are then evaluated through global feature rankings, local explanations, agreement, stability, local fidelity, and computation time. Explanations are analysed separately for true positives, true negatives, false positives, and false negatives. The paper defines the full experimental protocol and reporting structure. Results are intentionally left pending until the implementation produces verifiable measurements.

Index Terms—cybersecurity, explainable artificial intelligence, intrusion detection system, LIME, SHAP

I. INTRODUCTION

Intrusion detection systems examine network events for signs of misuse, attack, or policy violation. Machine-learning models extend this task beyond fixed signatures by learning statistical patterns from labelled traffic. This flexibility is valuable when attack behaviour varies, but it also creates an operational problem: a model may label a flow as malicious without giving the analyst a reason that can be checked against the traffic context.

An unexplained false positive can waste investigation time or encourage alert fatigue. An unexplained false negative is more serious because harmful activity may pass without review. Accuracy alone hides this difference. A useful evaluation must therefore report recall and false-positive rate, preserve the natural test distribution, and examine why correct and incorrect predictions receive their scores.

Explainable artificial intelligence provides methods for describing a trained model's behaviour. SHAP assigns contributions to input features through an additive framework derived from Shapley values [3]. LIME fits a simple surrogate around one prediction after creating nearby samples [2]. Both techniques can help expose influential attributes, but neither proves that a prediction is correct or that an attributed feature caused an attack. Explanations can also change with reference data, perturbation rules, or random seeds.

This paper proposes a controlled comparison of Random Forest and XGBoost on UNSW-NB15. It evaluates detection and explanation under one data protocol, then separates explanation analysis by true positive, true negative, false positive, and false negative outcomes. The central question is: How effectively can Explainable AI techniques improve the interpretability of machine-learning-based intrusion detection systems while maintaining reliable intrusion-detection performance?

II. BACKGROUND AND RELATED WORK

A. Explainability Methods

LIME explains a prediction by sampling around the selected record, obtaining model outputs for those samples, and fitting an interpretable local model [2]. Its model-agnostic design is useful when the same explanation procedure must be applied to different classifiers. The local approximation can nevertheless become unstable when sampling changes or when perturbations create implausible combinations of network attributes.

SHAP represents a prediction as a baseline plus feature contributions [3]. Tree-specific implementations can efficiently explain tree ensembles such as Random Forest and XGBoost. A SHAP summary can describe global model reliance, while waterfall

or bar plots can explain one alert. Attributions remain conditional on the model, feature representation, background sample, and dependence assumptions. Correlated traffic features can divide or redistribute importance in ways that require cautious interpretation.

B. Explainable Intrusion Detection

Wang et al. presented an early explainable machine-learning framework for IDS using SHAP with one-versus-all and multiclass classifiers on NSL-KDD [4]. The explanations corresponded to known attack characteristics, establishing a practical link between feature attribution and intrusion analysis. The use of an older benchmark and the limited treatment of explanation reliability leave room for evaluation on other traffic representations.

Patil et al. trained decision tree, Random Forest, SVM, and voting models on CICIDS2017 and used LIME to explain individual results [5]. Their study shows that ensemble detection and local explanation can be integrated. It concentrates on LIME and reported classifier performance, so agreement with another explainer and a systematic comparison of false positives and false negatives remain separate questions.

Wei et al. developed xNIDS to explain deep network intrusion models while accounting for input history and feature dependencies [6]. The framework evaluates fidelity, sparsity, completeness, and stability and can convert explanations into defence rules. This work demonstrates that explanation quality has measurable dimensions beyond visual appeal, although its deep-model setting differs from the tabular tree ensembles considered here.

Arreche et al. proposed XAI-IDS and evaluated seven classifiers with SHAP and LIME on CICIDS2017, NSL-KDD, and RoEduNet-SIMARGL2021 [7]. Their framework compares model behaviour, feature selection, and explanation techniques across datasets. It confirms that broad multi-model XAI comparisons already exist; a defensible contribution for the present project must therefore come from its controlled UNSW-NB15 protocol and error-stratified analysis.

Gaspar et al. applied LIME and SHAP to an MLP-based IoT intrusion solution using a generated form of the ADFA-LD system-call dataset [8]. They used perturbation analysis to test explanations and collected participant feedback. Their work supports explicit evaluation of explanations, while its system-call features and neural model limit direct transfer to network-flow tree models.

Mia et al. compared SHAP patterns for correct predictions and errors across CIC-IoT2023, NF-UQ-NIDS-v2, and HIKARI-2021 [9]. Their overlapping plots are designed to guide analysts in identifying potential false positives and false negatives. The approach gives the current study a direct methodological precedent;

the planned extension compares a second explainer, retains the original test distribution, and reports explanation time and stability.

Hermosilla et al. compared SHAP and LIME for XGBoost and TabNet on UNSW-NB15, including stability and fidelity considerations in a forensic setting [10]. This is the closest prior configuration and prevents a broad novelty claim based only on the dataset and explainers. The present study instead pairs XGBoost

with Random Forest and organises the comparison around all four confusion-matrix outcomes.

Mohale and Obagbuwa's systematic review found continued problems in standardisation, scalability, and real-time use of explanations in IDS [11]. Taken together, the literature shows that explanations should be evaluated as outputs with their own limitations. It also shows that explanation plots alone do not establish human understanding or operational trust.

Table I. Selected literature and its relevance to the proposed study

Study	Method	Dataset	Main finding	Limitation or gap
Wang et al. [4]	SHAP with neural IDS	NSL-KDD	Local and global attributions align with attack characteristics	Older dataset; limited explanation validation
Patil et al. [5]	Ensemble IDS with LIME	CICIDS2017	Local explanations accompany an ensemble classifier	Single explainer; limited error-stratified analysis
Wei et al. [6]	xNIDS	Four DL-NIDS settings	Measures fidelity, sparsity, completeness, and stability	Deep sequential setting differs from tabular tree models
Arreche et al. [7]	Seven models; SHAP and LIME	CICIDS2017, NSL-KDD, RoEduNet	Broad model and explainer comparison	Does not centre on UNSW-NB15 error outcomes
Gaspar et al. [8]	MLP; LIME and SHAP	Generated ADFA-LD	Perturbation and user feedback evaluate explanations	System-call data; one model family
Mia et al. [9]	SHAP error-group plots	Three traffic datasets	Diagnoses FP and FN patterns	No LIME comparison; visual similarity depends on analyst judgement
Hermosilla et al. [10]	XGBoost, TabNet; SHAP and LIME	UNSW-NB15	Compares fidelity and stability for forensic use	Different model pairing; broader outcome-stratified replication is useful
Mohale and Obagbuwa [11]	Systematic review	Twenty studies	Finds standardisation and scalability gaps	Review evidence does not replace controlled experiments

III. PROBLEM STATEMENT AND RESEARCH GAP

High-performing IDS classifiers can still be difficult to audit. A model comparison based only on accuracy cannot show whether one system misses more attacks, creates more false alarms, or relies on features that are unstable under a different explanation sample. Existing research has already applied SHAP and LIME to IDS, compared multiple models, and studied misclassifications [7]–[10]. The research gap is therefore specific rather than absolute.

The proposed contribution is a reproducible comparison of Random Forest and XGBoost under one UNSW-NB15 protocol. Detection metrics and explanation measures are reported together. Explanations are compared across the same records where possible and stratified by confusion-matrix outcome. The study also records agreement, stability, fidelity, and processing time. This design treats the work as a scoped replication and extension, not a new XAI algorithm.

IV. RESEARCH QUESTIONS AND OBJECTIVES

A. Research Questions

RQ1: How do Random Forest and XGBoost compare in attack recall, false-positive rate, F1-score, ROC-AUC, and PR-AUC under identical data partitions?

RQ2: Which original network attributes most strongly influence each model, and how much do their global feature rankings overlap?

RQ3: How do SHAP and LIME explanations differ among true positives, true negatives, false positives, and false negatives?

RQ4: How stable are the explanations under repeated LIME runs and controlled SHAP background samples, and what computation time do they require?

RQ5: Which diagnostic observations can be made from error explanations, and which claims about analyst trust remain untested without a user study?

B. Objectives

- Build a leakage-controlled preprocessing and training pipeline using the official UNSW-NB15 partitions.
- Compare Random Forest and XGBoost using the same test records and clearly defined positive class.

- Generate global and local explanations with network attributes presented in their original semantic groups.
- Analyse correct predictions, false alarms, and missed attacks, then measure explanation agreement, sensitivity, fidelity, and time.

V. PROPOSED METHODOLOGY

The study follows this sequence: dataset acquisition, data audit, preprocessing, training-only feature selection, model tuning, intrusion prediction, performance evaluation, SHAP and LIME explanation, error-group analysis, model comparison, and conclusion. A versioned notebook should reproduce every metric, figure, and selected case identifier.

A. Dataset Description

UNSW-NB15 was generated in the Cyber Range Lab at UNSW Canberra using IXIA PerfectStorm to combine normal activity with synthetic contemporary attacks [1]. The complete collection contains 2,540,044 records and 49 attributes including the class label. Its published training and test files contain 175,341 and 82,332 records, respectively. Attack categories include Analysis, Backdoors, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, and Worms [1].

The primary task is binary classification, with attack as the positive class. The attack category is excluded from predictors and retained only for error reporting. The fixed test partition supports repeatability, while its laboratory origin and age restrict claims about modern production traffic.

Table II. UNSW-NB15 study profile

Property	Planned use
Training records	175,341; transformations and model selection only
Test records	82,332; untouched final evaluation
Prediction task	Benign versus attack
Attack category	Evaluation metadata for missed-attack analysis
Feature types	Numeric and categorical network-flow attributes

B. Data Preprocessing and Feature Selection

The implementation will first record file checksums, schema, missing values, class counts, duplicates, and possible overlap

between the official partitions. Identifiers, binary label, and attack-category label will not enter the predictor matrix. Invalid numeric values will be converted to missing values and handled using a training-fitted rule. Protocol, service, and state will be encoded with an explicit policy for categories that appear only in the test set.

Constant attributes will be removed. Highly redundant features will be identified through training data and domain review. If supervised feature selection is used, it will be fitted within training folds and applied identically to both models. A full-feature baseline will show whether selection changes detection quality. Resampling, if required, will occur only in training data. The original test distribution will remain unchanged because balancing it would distort false-positive and prevalence-sensitive results.

C. Machine Learning Models

Random Forest builds an ensemble of decision trees from bootstrapped samples and random feature subsets [12]. Averaging across trees can reduce the variance of a single tree, while class weights and depth controls help manage imbalance and complexity. XGBoost adds trees sequentially to correct current errors and uses regularisation, shrinkage, and sampling to control overfitting [13]. Both models support probability estimates and tree-based SHAP, which permits a consistent explanation interface.

A validation subset within the training partition will guide hyperparameter selection. Candidate settings will cover tree count, depth, minimum leaf or child weight, feature sampling, learning rate, and regularisation as applicable. Both models will receive comparable tuning effort. Random seeds, final settings, software versions, and hardware will be recorded.

D. Explainable AI Techniques

TreeSHAP will produce global mean absolute attributions and local explanations for selected records. The analysis will record the background sample, class output, and whether contributions explain raw scores or probabilities. One-hot features will be aggregated back to original attributes so that a categorical variable is interpreted as one concept rather than several unrelated columns.

LIME will call each trained model through the same preprocessing pipeline. Its perturbation process will treat categorical variables as categories rather than independently switching one-hot indicators. Multiple random seeds will be used for selected cases. Local surrogate fit will be recorded because a low-fidelity explanation should not be treated as a close description of the original classifier.

E. Experimental Setup

The official test file is reserved for final evaluation. Model and preprocessing choices are made with three stratified folds or a fixed validation subset drawn only from the training file. If the data audit identifies duplicate or related records, the split strategy will group them where practical to reduce information leakage. The default decision threshold is 0.5. Any alternative threshold must be chosen on validation data against a declared false-alarm objective before test evaluation.

Global SHAP summaries will use a representative test sample. For local analysis, a reproducible sample of up to 50 records from each of the four outcomes will be selected for each model, using all records when fewer are available. The same record identifiers will be compared across models and explainers where feasible, even when models place a record in different outcome groups.

F. Evaluation Metrics

Accuracy is the fraction of correct decisions. Precision measures the fraction of predicted attacks that are attacks. Recall measures the fraction of attacks detected. F1 is the harmonic mean of precision and recall. False-positive rate is $FP/(FP+TN)$. ROC-AUC evaluates

ranking across thresholds, while PR-AUC is added because it is more informative when class prevalence is unequal. Confusion matrices and attack-category recall will make errors visible.

Explanation comparison will include top-five Jaccard overlap, rank agreement where meaningful, LIME local surrogate fidelity, attribution stability across repeated settings, and median time per explained record. These measures answer different questions. Agreement between SHAP and LIME does not prove that either explanation is true, and a stable explanation does not prove the underlying prediction is correct.

VI. RESULTS AND ANALYSIS

To be updated after implementation. The tables below define the results that will be reported. Pending cells do not represent estimates or expected performance. Each table will include the evaluated sample count and the final decision threshold.

Table III. Planned model performance report

Model	Prec.	Recall	F1	FPR	ROC-AUC	PR-AUC
Random Forest	Pending	Pending	Pending	Pending	Pending	Pending
XGBoost	Pending	Pending	Pending	Pending	Pending	Pending

The final analysis will compare recall and false-positive rate before using aggregate accuracy to choose a preferred model. ROC and precision-recall curves will show threshold behaviour. Confusion matrices and category-level attack recall will identify whether a strong overall score hides weak detection of rare attacks.

A. Explainability Analysis

To be updated after implementation. Global SHAP summaries will identify the attributes most strongly associated with model output, while local SHAP and LIME panels will show how individual values support or oppose a prediction. The paper will report whether the two models rely on similar attributes and whether SHAP and LIME agree on their leading local features. Explanations will use feature descriptions grounded in the UNSW-NB15 schema rather than treating column names as self-explanatory.

Table IV. Planned explanation comparison

Model	Explainer	Outcome	n	Agreement	Stability	Time
RF	SHAP	TP/TN/FP/FN	Pending	Pending	Pending	Pending
RF	LIME	TP/TN/FP/FN	Pending	Pending	Pending	Pending
XGB	SHAP	TP/TN/FP/FN	Pending	Pending	Pending	Pending
XGB	LIME	TP/TN/FP/FN	Pending	Pending	Pending	Pending

B. False Positive and False Negative Analysis

To be updated after implementation. False positives will be compared with true positives to identify benign records whose values resemble the model's attack pattern. False negatives will be compared with true negatives and separated by attack category to identify attacks that resemble benign traffic. Each case study will report the true label, predicted label, probability, leading positive and negative contributions, agreement between explainers, and a restrained interpretation.

The analysis will avoid causal language. For example, a high SHAP value for a traffic-rate feature means that the model used that observed value to shift its output; it does not prove that the feature caused the attack. Cases with low LIME fidelity, unstable feature rankings, or unrealistic perturbations will be marked as uncertain rather than forced into a narrative.

VII. DISCUSSION

The completed study should be interpreted on two levels. Detection metrics describe how the trained classifier behaves on the fixed benchmark. Explanation metrics describe how consistently selected XAI methods approximate or attribute that behaviour. A model may achieve higher recall but rely on unstable explanations, or an explainer may provide concise output while imposing unacceptable computation time. The final discussion will keep these findings separate.

Post-hoc explanation does not change a fixed model's predictions. Adding SHAP or LIME therefore cannot directly improve accuracy, recall, or false-positive rate. Explanations may support later model redesign, feature review, or threshold adjustment, but each change requires validation on data that did not inform the change. Once test cases guide redesign, those cases are no longer an untouched final evaluation set.

Claims about trust and practical usability require evidence from security analysts. The present benchmark can demonstrate feature visibility, explanation consistency, and processing cost. It cannot demonstrate faster triage, better analyst decisions, or greater trust without a user study. This boundary is important because readable plots may still omit dependencies or reflect artefacts in the data.

VIII. LIMITATIONS

UNSW-NB15 contains laboratory-generated traffic and may not represent present production networks, encrypted applications, or current attack behaviour. Its predefined split supports repeatability but does not test temporal generalisation or cross-network transfer. Binary labels merge distinct attacks, and rare categories may yield too few errors for stable subgroup analysis.

SHAP and LIME explain model behaviour rather than network causality. Correlated features, one-hot encoding, background selection, local sampling, and probability calibration can alter attribution. LIME may generate improbable neighbourhoods; SHAP results depend on the selected explainer and reference distribution. Sampling explanations reduces cost but introduces uncertainty. Finally, no analyst study is planned in the initial implementation, so human usefulness will remain a proposed future evaluation.

IX. FUTURE SCOPE

After the binary benchmark is complete, the protocol can be extended to multiclass attack recognition, temporal validation, and cross-dataset testing on CICIDS2017 or a newer operational collection. Counterfactual explanations could complement feature attribution by showing feasible changes associated with a different decision. A user study with security analysts could measure decision accuracy, review time, confidence calibration, and the effect of explanation uncertainty. Runtime profiling could also test selective explanation of high-risk alerts rather than every flow.

X. CONCLUSION

This paper defines a reproducible study of explainable machine-learning decisions in network intrusion detection. Random Forest and XGBoost will be evaluated on the official UNSW-NB15 partitions with leakage-controlled preprocessing and metrics that distinguish missed attacks from false alarms. SHAP and LIME will be compared through global and local attributions, correct and incorrect prediction groups, agreement, stability, fidelity, and computation time. The design reflects prior work and presents a focused replication and extension rather than an unsupported novelty claim. Experimental conclusions will be added only after the implementation produces verifiable results.

REFERENCES

- [1] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in 2015 Military Communications and Information Systems Conference (MilCIS), 2015, pp. 1–6, doi: 10.1109/MilCIS.2015.7348942.
- [2] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 1135–1144, doi: 10.1145/2939672.2939778.
- [3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 4765–4774.
- [4] M. Wang, K. Zheng, Y. Yang, and X. Wang, "An explainable machine learning framework for intrusion detection systems," IEEE Access, vol. 8, pp. 73127–73141, 2020, doi: 10.1109/ACCESS.2020.2988359.
- [5] S. Patil et al., "Explainable artificial intelligence for intrusion detection system," Electronics, vol. 11, no. 19, Art. no. 3079, 2022, doi: 10.3390/electronics11193079.
- [6] F. Wei, H. Li, Z. Zhao, and H. Hu, "xNIDS: Explaining deep learning-based network intrusion detection systems for active intrusion responses," in 32nd USENIX Security Symposium, 2023, pp. 4337–4354.
- [7] O. Arreche, T. Guntur, and M. Abdallah, "XAI-IDS: Toward proposing an explainable artificial intelligence framework for enhancing network intrusion detection systems," Applied Sciences, vol. 14, no. 10, Art. no. 4170, 2024, doi: 10.3390/app14104170.
- [8] D. Gaspar, P. Silva, and C. Silva, "Explainable AI for intrusion detection systems: LIME and SHAP applicability on multi-layer perceptron," IEEE Access, vol. 12, pp. 30164–30175, 2024, doi: 10.1109/ACCESS.2024.3368377.
- [9] M. Mia, M. M. A. Pritom, T. Islam, and K. Hasan, "Visually analyze SHAP plots to diagnose misclassifications in ML-based intrusion detection," in 2024 IEEE International Conference on Data Mining Workshops, 2024, pp. 632–641, doi: 10.1109/ICDMW65004.2024.00088.
- [10] P. Hermosilla, S. Berrios, and H. Allende-Cid, "Explainable AI for forensic analysis: A comparative study of SHAP and LIME in intrusion detection models," Applied Sciences, vol. 15, no. 13, Art. no. 7329, 2025, doi: 10.3390/app15137329.
- [11] V. Z. Mohale and I. C. Obagbuwa, "A systematic review on the integration of explainable artificial intelligence in intrusion detection systems to enhancing transparency and interpretability in cybersecurity," Frontiers in Artificial Intelligence, vol. 8, Art. no. 1526221, 2025, doi: 10.3389/frai.2025.1526221.
- [12] L. Breiman, "Random forests," Machine Learning, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.
- [13] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.